MD. MEHERAB HOSSEN

Mechanical Engineer | CFD & Thermal Sciences

Dhaka–1217, Bangladesh | +880 1538-280009 | meherabhossen3085@gmail.com

meherabhossen.pages.dev | GitHub: cfdmeherabhossen | ResearchGate

Research Profile

I am a Mechanical Engineering graduate and CFD Simulation Engineer working on multiphase flow, heat transfer, and phase change. I develop and validate numerical models in ANSYS Fluent and OpenFOAM against published benchmarks, with research in thermal energy storage and droplet dynamics.

Education

Ahsanullah University of Science and Technology **April 2022 – July 2026**
B.Sc. in Mechanical Engineering CGPA: 3.122/4.00

Research Interests

Computational multiphase flow; droplet impact and heat transfer; phase-change thermal energy storage; benchmark-based numerical model validation.

Manuscripts & Conferences

Md. Meherab Hossen, Md. Asif Farhan, Tabasom Rahman, Hasibuzzaman Mahmud, Dewan Hasan Ahmed. "Numerical Investigation of Enhanced Thermal Energy Storage in Solid-Solid Phase Change Materials with New Fin Architecture for Shell-and-Tube Heat Exchanger Applications."
Journal of Energy Storage. First author; submitted October 2026; **under review**.

10th BSME International Conference on Thermo-Fluid Engineering
Abstracts accepted; full papers under review (submitted September 2026).

1. **Md. Meherab Hossen**, Md. Asif Farhan, Radia Siddika, Nowroze Farhan Ifraj. "Numerical Investigation of the Coupled Effect of Impact Angle and Ambient Rotor Downwash on Rain-Droplet Impact Regimes over UAV Surface." *First author.*
2. Md. Asif Farhan, **Md. Meherab Hossen**, Radia Siddika, Nowroze Farhan Ifraj. "Effect of Pre-Impact Thermal History on the Impact and Freezing of Free-Falling Supercooled Droplets on UAV-Relevant Surfaces." *Second author.*
3. Tabasom Rahman, Md. Asif Farhan, **Md. Meherab Hossen**, Fahmid Zaman Adnan, Ibrarur Rahman Choudhury, Dewan Hasan Ahmed. "Thermal Energy Storage Enhancement Using Alphabet-Inspired Fins in a Shell-and-Tube Heat Exchanger with Solid–Liquid PCM." *Third author.*

Research Experience

[Validation figures and simulation portfolio]

Undergraduate Research Assistant – Thermal Energy Storage **2025 – 2026**
Advisor: Prof. Dr. Dewan Hasan Ahmed, Ahsanullah University of Science and Technology

- Developed Fluent phase-change models using temperature-dependent apparent heat capacity for the solid–solid transition at 25 °C; designed bio-inspired straight-fin arrangements.
- Validated liquid-fraction histories against Alnakeeb et al. and temperature histories against Farulla et al.; contributed to mesh checks using fully melted time in the solid–liquid PCM study.
- Analyzed five fin angles and orientations against an unfinned baseline in the solid–solid PCM study; the best simulated configuration reduced charging time by 79.3%.

Research Experience (continued)

Research Assistant – Droplet Dynamics **2026 – Present**

Research mentor: Nowroze Farhan Ifraj, PhD student in Mechanical Engineering, Virginia Tech

- Configure VOF droplet models and analyze Weber-number effects on spreading and deformation.
- Reproduced Rafi et al. (2022) falling-droplet deformation and collision cases in ANSYS Fluent, with a spread-ratio error of 3%. [Validation]
- Benchmarked OpenFOAM droplet-pair impact simulations against Goswami and Hardalupas (2023) over We = 54–128, with an 8% deviation from published data. [Validation] [Videos]

Professional Experience

CFD Simulation Engineer **July 2026 – Present**

MACPROTEC, Houston, Texas, USA | Remote

- Analyze industrial dust collectors, baghouses and process pipelines using ANSYS Fluent, ANSYS Rocky and OpenFOAM.
- Develop 3D geometry in ANSYS DesignModeler and SolidWorks for simulation workflows.
- Reconstruct CAD geometry from laser scans using FARO SCENE, CloudCompare, MeshLab and Geomagic Design X.
- Configure discrete phase model (DPM) simulations of air–coal and air–dust particle transport.

Thesis

Experimental Characterization of *Nymphaea rubra* (Water Lily) Natural Fiber and Its Epoxy Composite through Mechanical Analysis **July 2026**

Supervisor: Dr. Fazle Rabbi, Ahsanullah University of Science and Technology | Three-member team

- Investigated water-lily fiber as a sustainable reinforcement for epoxy composites; conducted ASTM-based single-fiber tensile testing using a universal testing machine.
- Identified 48-hour retting as the best extraction condition. Composite properties: tensile strength 25.5 ± 2 MPa; elastic modulus 3.2 ± 0.29 GPa; flexural strength 22 ± 5.5 MPa. [Research] [Lab gallery]

Projects

- **NACA 23012 Airfoil.** Analyzed the airfoil at $\mathrm{Re} = 2 \times 10^6$ over angles of attack from -5° to 24°; compared results with Miley's (1982) wind-tunnel data. *Tools: XFLR5, Microsoft Excel.*
- **50 kW Grid-Connected PV.** Estimated 76,733 kWh/year and a 3.26-year payback. [Report]

Skills

CFD & simulation: ANSYS Fluent, OpenFOAM, ANSYS Rocky, COMSOL.
CAD & reverse engineering: SolidWorks, ANSYS DesignModeler, FARO SCENE, CloudCompare, Geomagic Design X, MeshLab.
Programming: C, C++, MATLAB, Python.
Scientific communication: LaTeX, Microsoft Word, Excel (advanced), PowerPoint.

Awards & Training

- BITAC training in CNC/lathe machining, heat treatment, molding, and casting. [Certificate]
- **People's Choice Award – Mechanical Project Showcasing, Fall 2023.** [Portfolio]
- **MATLAB Onramp – MathWorks.** [Certificate]

References

Prof. Dr. Dewan Hasan Ahmed
Department of Mechanical & Production Engineering, Ahsanullah University of Science and Technology
dhahmed.mpe@aust.edu

Dr. Fazle Rabbi
Associate Professor, Department of Mechanical Engineering, Ahsanullah University of Science and Technology
fazle.rabbi.mpe@aust.edu